

Machine Translation of Public Service Announcements in Kenya: English / Kiswahili ↔ Ekegusii

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INTRODUCTION

- **Problem:** Kenyan government Public Service Announcements (PSAs) — health, security, education, agriculture, governance — rarely reach citizens in indigenous languages.
- **Ekegusii** (Gusii region, ~2.7M speakers) is under-resourced: no large parallel corpora, and absent from major multilingual MT models' training sets.
- **Mission:** build and evaluate a proof-of-concept MT system that translates PSAs between English/Kiswahili and Ekegusii via few-shot cross-lingual transfer — and ship it as a working demo.

PRIOR WORK

- **NLLB-200** (Meta) covers 204 languages — Ekegusii is **not one of them**; its tokenizer has no *guz_Latn* token.
- **FLORES-200**, the standard low-resource benchmark, was evaluated and **dropped: it contains no Ekegusii**. We built our own 138-pair held-out benchmark instead.
- **mT5** and NLLB show strong multilingual transfer for high-resource pairs; few-shot transfer to *unseen* languages remains the open challenge we target.
- **TICO-19** (CC BY 4.0) provides professional COVID-era EN–SW translation memory — our largest corpus source.

METHODOLOGY

Table 1 — PSA dataset — 6,823 rows, five brief domains

Source tier	Rows
Lecturer gold (EN/SW/ GUZ /Dholuo/Somali)	2,852
TICO-19 corpus (EN–SW)	3,004
Audited scraped PSAs (21 sites)	817
Team-written PSAs	150

- **Data pipeline:** robots.txt-respecting scraping → lecturer-calibrated **PSA framework audit** (9,548 off-framework rows deleted) → cleaning, mojibake repair → stratified 90/5/5 splits (6,141/341/341, zero leakage).
- **Model:** NLLB-200-600M fine-tuned on a Navon Cloud/Kinesis **A100-80GB**; a new *guz_Latn* token added with embeddings **donor-initialized from *swh_Latn***.
- **Ablations** (9 runs): mT5-small vs NLLB, freezing, 50/200/all guz pairs — tracked in Weights & Biases.
- **Augmentation:** eng → guz back-translation of 3,577 guz-less train rows by our own model (loop-guarded decoding, copy-through filtered).

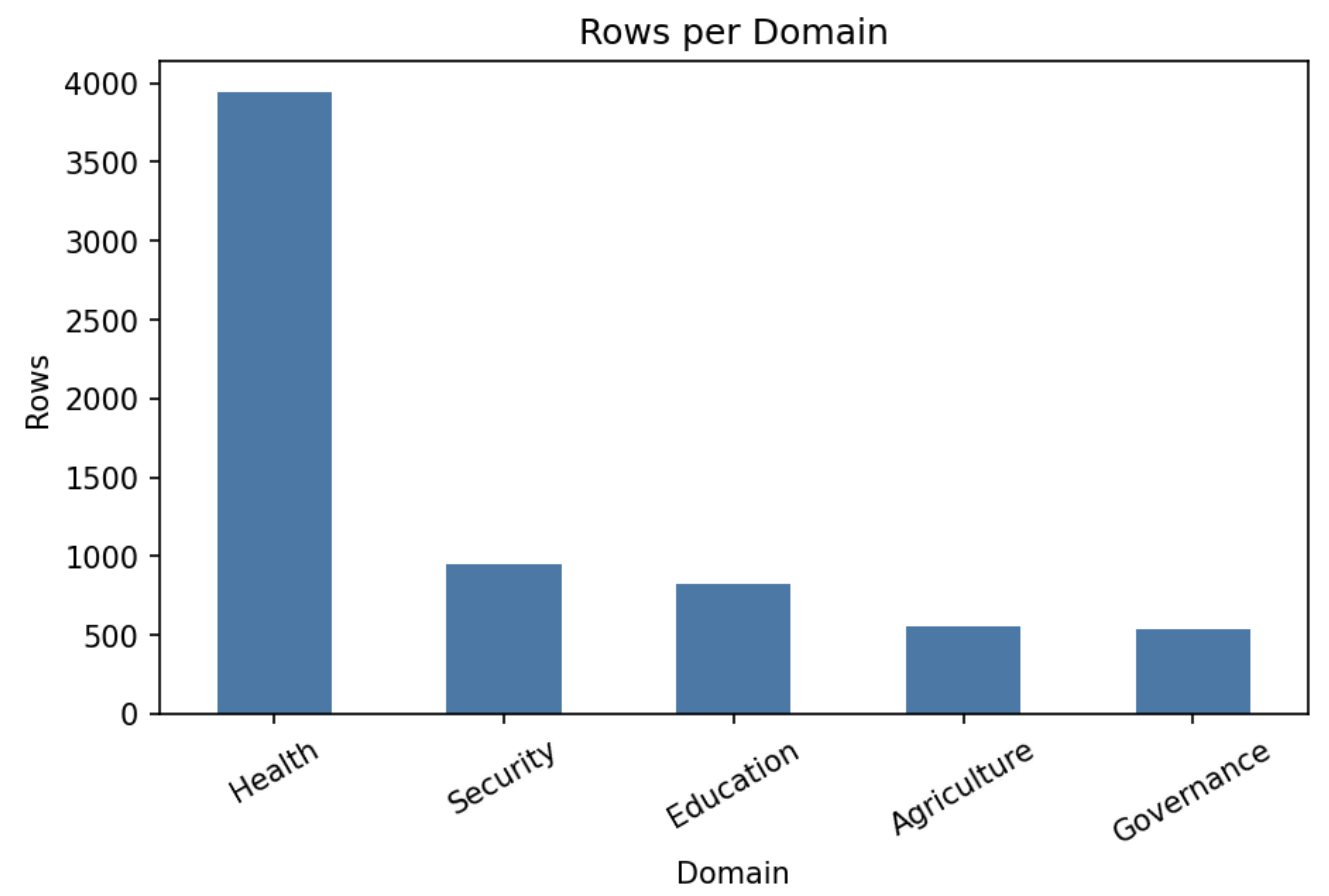


Figure 1. PSA rows per domain after the framework audit. Health dominates (57.8%) — down from 85% mid-pipeline — with training done on a domain-balanced view.

RESULTS — ENGLISH ↔ KISWAHILI

Table 2 — Full held-out test split (304 pairs, NLLB ft)

Configuration	BLEU	chrF
Zero-shot NLLB-600M	5.57	—
Fine-tuned (dev)	50.93	72.34
Fine-tuned (test, en → sw)	49.31	72.23
Fine-tuned (test, sw → en)	48.38	69.10

Domain adaptation is worth **+45 BLEU over zero-shot**; test-set drift vs dev is only 0.25 BLEU — the model genuinely generalizes.

RESULTS — EKEGUSII TRANSFER

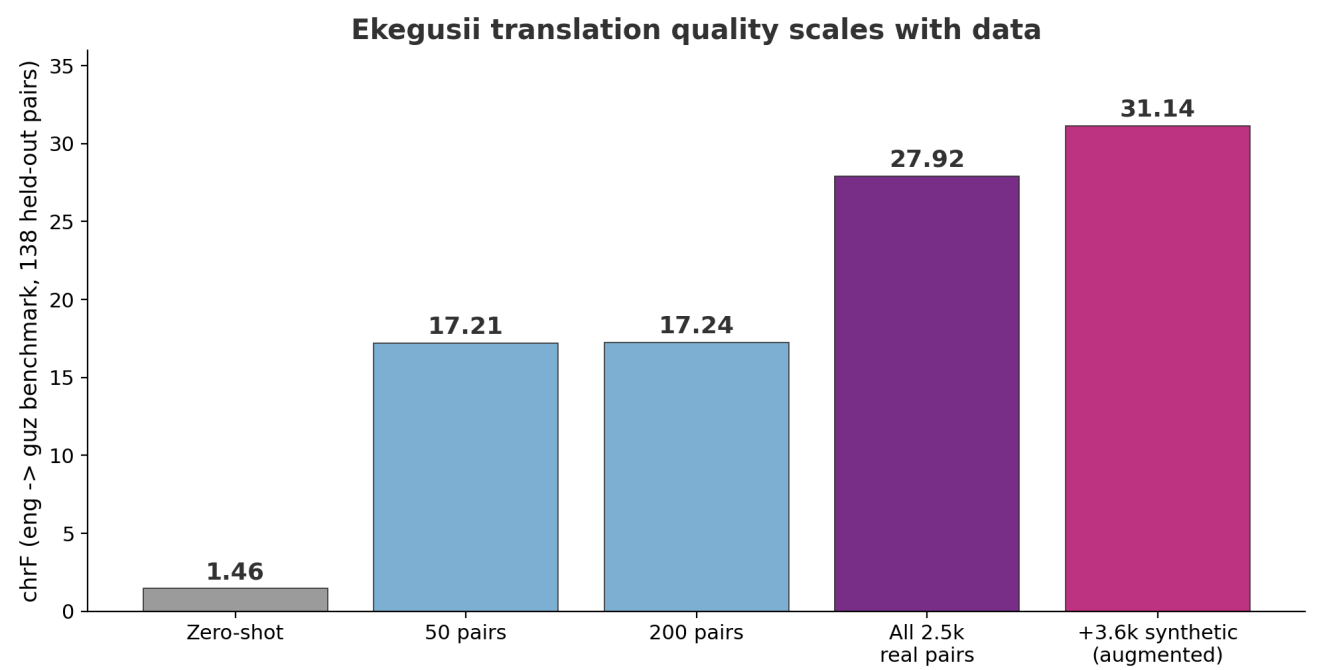


Figure 2. With 200 pairs the model outputs fluent Swahili when asked for Ekegusii (donor-language confusion); real Ekegusii morphology emerges with the full gold set, and back-translation augmentation lifts it further.

Table 3 — Augmentation experiment (en → guz, 138-pair benchmark)

Metric	Real only	+ Synthetic
chrF	27.92	31.14
BLEU	3.56	4.58
Repetition loops	23/138 (16.7%)	14/138 (10.1%)

Table 4 — en → guz chrF by PSA domain (benchmark, pre-augmentation)

Domain	n	chrF	Loops
Governance	18	34.16	1
Education	38	29.30	3
Agriculture	19	28.34	3
Security	38	26.15	8
Health	25	24.30	8

Short, formulaic PSA language transfers best; repetition loops concentrate in the two lowest-scoring domains.

INTERPRETATION

- Few-shot transfer to an **unseen language works**: donor-initialized tokens + ~2.5k gold pairs yield grammatical Ekegusii for most PSA inputs.
- Data volume — not model size — is the bottleneck: the controlled augmentation experiment converts that hypothesis into a **measured +3.2 chrF / –39% loops**.
- Error analysis is quantified, not anecdotal: loops and language confusion are counted per domain; guz → en (chrF 20.1) remains the hardest direction.

LIVE DEMO (STREAMLIT APP)

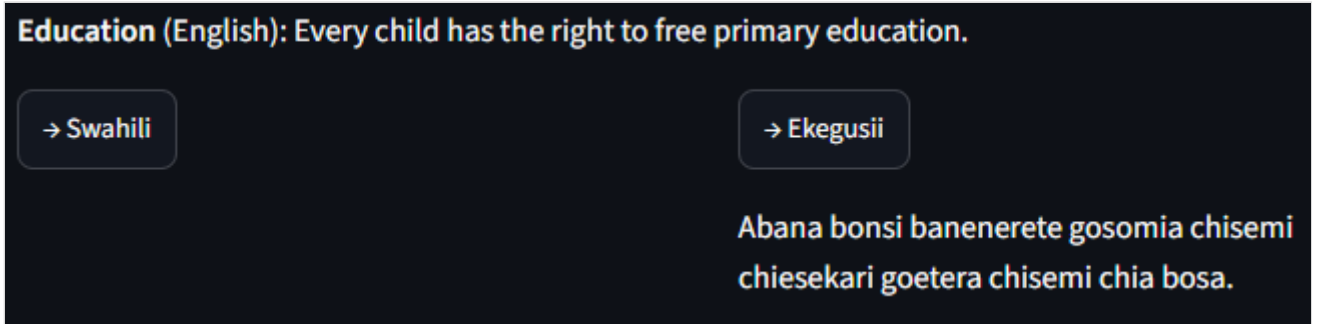


Figure 3. en → guz: “Every child has the right to free primary education.” → “Omwana ense nigo enenerete gosomera chisukuru chioboroso.”

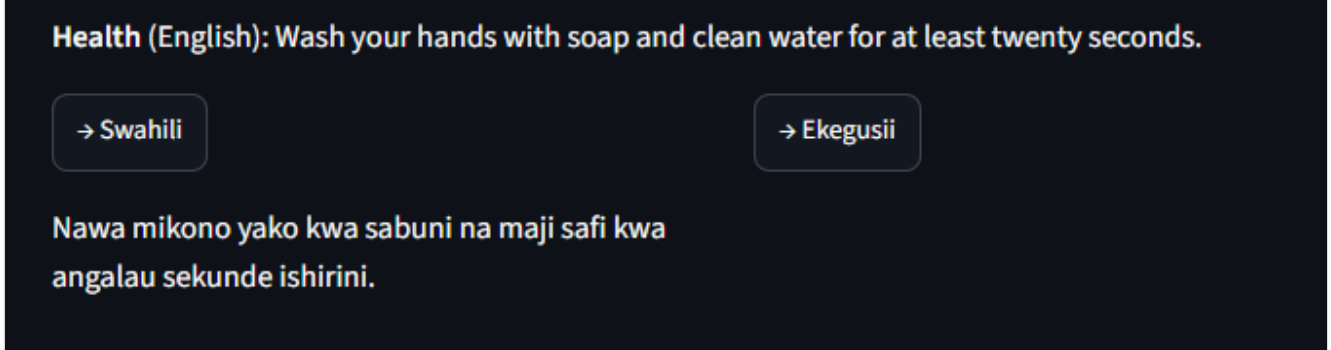


Figure 4. en → sw: hand-washing PSA translated with domain-appropriate register. The app wraps the fine-tuned checkpoint with loop-guarded decoding.

CHALLENGES & LIMITATIONS

- **Tiny gold corpus:** ~2.5k Ekegusii pairs demonstrate transfer, not native fluency.
- **Residual failure modes** (reduced, not eliminated): repetition loops on long inputs, rare language confusion / copy-through on low-confidence inputs.
- **Self-training ceiling:** synthetic data consolidates what the model knows; the next real gain needs more *human* Ekegusii translation.
- **Domain skew:** Health is 57.8% of data, and its clinical terminology is the weakest EN–SW subdomain (44.8 BLEU) despite its volume.

FUTURE WORK

- **Native-speaker human evaluation** of model Ekegusii (fluency/adequacy 1–5) — 30-row stratified sheet and reviewer guide already prepared.
- **More human Ekegusii data** via community/University partnerships; extend augmentation with vetted monolingual text.
- **guz → en improvements** (larger checkpoint, direction-balanced training) and decoding-time terminology control with the 24-term Kenyan institutional glossary.
- **Public deployment** of the web app for county governments and NGOs.

Takeaway: with a donor-initialized language token, disciplined data curation, and one measured augmentation step, a 600M-parameter model can bring an unseen Kenyan language into a working PSA translation system.